CS & IT ENGINEERING

Discrete Mathematics

Mathematical Logic

DPP 01 Discussion Notes



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[MCQ]

#Q. Which of the following is tautology?

A) $(\sim p \wedge (p \to q)) \to \sim q$

B) $\sim (p \to q)) \to \sim q$

C) $[(\sim p \wedge q) \wedge (q \to (p \to q))] \to \sim r$

D) None of these

A) $(\sim p \wedge (p \to q)) \to \sim q$

$\equiv (\sim p \wedge (\sim p \vee q)) \to \sim q$

By absorption law $(p \vee (p \wedge q) \equiv p$
$p \wedge (p \vee q) \equiv p)$

$\equiv \sim p \to \sim q$

B) $\sim (p \to q) \to \sim q$

$\equiv (p \wedge \sim q) \to \sim q$

$\equiv \sim p \vee q \vee \sim q$

$\equiv T$

$$((\sim p \wedge q) \wedge (q \rightarrow (p \rightarrow q))) \rightarrow \sim r$$

$$\equiv ((\sim p \wedge q) \wedge \underbrace{(\sim q \vee \sim p \vee q)}_{T}) \rightarrow \sim r$$

$$\equiv (\sim p \wedge q) \rightarrow \sim r$$

[MCQ]

#Q. The statement [PV (p ↔ Q) V Q] is equivalent to

$p \vee (p \leftrightarrow q) \vee q \equiv T$

$\equiv$

A P

B Q

C A tautology

D $(P \wedge Q)$

[MSQ]

#Q. Consider the following statement

S_1: $[(p \Rightarrow q) \wedge (q \Rightarrow r)] \Rightarrow (r \Rightarrow p)$

S_2: $[(p \Rightarrow q) \wedge (q \Rightarrow r)] \Rightarrow (p \Rightarrow r)$

A. S_1 is contingency

B. S_2 is tautology

C. S_1 and S_2 both contingency

D. S_1 and S_2 both Tautology

$S_1 \equiv ((p \to q) \wedge (q \to r)) \to (r \to p)$

$p \to q$, $q \to r$ — Transitive

$\therefore p \to r$

$p \to r \not\equiv r \to p$

[MCQ]

#Q. Which of the following is valid?

S_1: $p \Rightarrow (q \vee r) \equiv (p \Rightarrow q) \vee (p \Rightarrow r)]$

S_2: $p \Rightarrow (q \wedge r) \equiv (p \Rightarrow q) \wedge (p \Rightarrow r)]$

A S_1 is valid and S_2 is not valid

B S_2 is not valid and S_2 is valid

C Both S_1 and S_2 are valid

D Neither S_1 and S_2 is valid

S_1: LHS:

$p \to (q \vee r) \equiv \sim p \vee q \vee r$

RHS:

$(p \to q) \vee (p \to r)$

$\equiv \sim p \vee q \vee \sim p \vee r$

$= \sim p \vee q \vee r$

S_2: LHS

$p \to q \wedge r = \sim p \vee (q \wedge r)$

RHS :

$$(p \to q) \wedge (p \to r)$$

$$\equiv (\sim p \vee q) \wedge (\sim p \vee r)$$

$$\equiv \sim p \vee (q \wedge r)$$

[MSQ]

#Q. Which of the following is/are true.

A) $\{R \to S, P \to Q, R \vee P\} \Rightarrow S \vee Q$

$R \to S$
$P \to Q$
$\to R \vee P$
$S \vee Q$

Constructive dilemma

B) $\{\sim R \to (S \to \sim T), \sim R \vee W, \sim P \to S, \sim W\} \Rightarrow T \to P$

$\sim R \to (S \to \sim T)$ (T, T)
$T \quad \sim R \vee W$ (F)
$\sim\sim P \to S$
$T \quad \sim W$
$T \to P$ ✓

C) $\{\sim P \wedge Q, Q \to (P \to R)\} \Rightarrow \sim R$

$S \to \sim T$
$\sim P \to S$

$\equiv \sim S \vee \sim T$
$P \vee S$

$\equiv \sim S \vee \sim T$
$S \vee P$

Resolution

D) $\{P \to R, P, Q \vee \sim R\} \Rightarrow Q$

D) $P \to R$: T

T : P

$Q \vee \sim R$: F

$\sim Q$

$\sim T \vee P$

$\Rightarrow T \to P$

[MCQ]

#Q. Consider the following logical inferences.

I_1 : "I will study discrete math or I will study English literature"
"I will not study discrete math"
Inference : I will not study English literature

I_2 : "If A works hard then B or C will enjoy themselves"
"If B enjoys himself then A will not work hard"
"If C enjoys himself then D will not enjoy himself"
Inference: - If A work hard then D will not enjoy himself

I_1 : p : I will study DM

Q : I will study Eng

1°: $p \vee q$ (F, T)

2. $\sim p$ (T)

$\sim q$ — q

A. Both I_1 and I_2 are correct inferences

B. I_1 is correct but I_2 is not a correct inference

C. I_1 is not correct but I_2 is a correct inference

D. Both I_1 and I_2 are not correct inferences

I_2 :

A : A works Land

B : B enjoys himself

C : C " "

D : D "

1. $A^{T} \rightarrow (B^{F} \vee C^{F}) \rightarrow F$
2. $B^{F} \rightarrow \sim A^{F} \rightarrow T$
3. $C^{F} \rightarrow \sim D^{F} \rightarrow T$

$A^{T} \rightarrow \sim D^{F} \rightarrow F$

$\rightarrow F$

[MSQ]

#Q. Which of the following is/are logical equivalence?

I. $\sim (p \to q)$ — $p \wedge \sim q$

II. $\sim (p \to q) \wedge (q \to r)$

III. $p \wedge \sim q$

IV. $(p \vee q) \to r$

$(p \wedge \sim q) \wedge (\sim q \vee r)$

$\equiv (p \wedge \sim q \wedge \sim q) \vee (p \wedge \sim q \wedge r)$

$= (p \wedge \sim q) \vee (p \wedge \sim q \wedge r)$

$\equiv (p \wedge \sim q)$

A I and II

B I and III

C II and IV

D II and III

$(p \vee q) \to r$

$\equiv (\sim p \wedge \sim q) \vee r$

[MCQ]

#Q. Consider the following statement

$S_1 : \sim (p \leftrightarrow q)$

$S_2 : p \leftrightarrow \sim q$

Which of the following is correct?

A S_1 is tautology

B S_2 is ~~concentration~~ contradiction

C S_1 is equivalence to S_2

D None of these

$S_1 : \sim (p \leftrightarrow q)$

$\equiv (\sim p \wedge q) \vee (p \wedge \sim q)$ XOR

$S_2 : p \leftrightarrow \sim q$

$\equiv (p \wedge \sim q) \vee (\sim p \wedge q)$

$p \Leftrightarrow q$

$(p \wedge q) \vee (\sim p \wedge \sim q)$

[MCQ]

#Q. Consider the following statement

$S_1 : \sim (p \vee (\sim p \wedge q))$

$S_2 : \sim p \wedge \sim q$

Which of the following is correct?

A S_1 is tautology

B S_2 is ~~concentration~~ contradiction

C S_1 is equivalence to S_2

D S_1 is not equivalence to S_2

$S_1 : \sim (p \vee (\sim p \wedge q))$

$\sim p \wedge \sim (\sim p \wedge q)$

$\sim p \wedge (p \vee \sim q)$

$(\sim p \wedge p) \vee (\sim p \wedge \sim q)$

F

$\equiv \sim p \wedge \sim q$

[MCQ]

#Q. $(p \to (q \to r)) \leftrightarrow ((p \to q) \to r)$ is

Associative

- True — all are True
- False — all are False

A Tautology

B Contingency

C Contradiction

D Can's be determined

THANK - YOU